From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

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Links

Research Compendium

The data and materials for this study are available on OSF (https://osf.io/uenr7) and GitHub (https://github.com/jeksterslab/manMetaVAR, https://jeksterslab.github.io/manMetaVAR/index.html).

fitVARMxID R Package

Source code and documentation for the `fitVARMxID` `R` package are available on GitHub (https://github.com/jeksterslab/fitVARMxID, https://jeksterslab.github.io/fitVARMxID/index.html).

metaDyn R Package

Source code and documentation for the `metaDyn` `R` package are available on GitHub (https://github.com/jeksterslab/metaDyn, https://jeksterslab.github.io/metaDyn/index.html).

Single Replication from the Simulation Study

This vignette presents a complete walk-through of one representative Monte Carlo replication, including data generation, estimation for each competing method, and selected output and diagnostics. Convergence diagnostics for a representative BMLVAR (Bayesian multilevel vector autoregressive model) replication, including PSR, ESS, and trace plots, are provided via the `Mplus` output section of the simulation-replication vignette.

https://jeksterslab.github.io/manMetaVAR/articles/sim-rep.html

Empirical Illustration: Meta-Analysis of Affect Dynamics

This vignette reproduces the empirical example reported in the manuscript, including data preprocessing, person-specific VAR estimation, second-stage meta-analytic synthesis, and the resulting population-level affect-dynamics summaries.

https://jeksterslab.github.io/manMetaVAR/articles/empirical.html

Benchmarking MetaVAR Against an Uncertainty-Uncorrected Two-Stage Approach and Bayesian Multilevel VAR

This vignette reports the benchmarking analysis comparing the computational demands of MetaVAR, the uncertainty-uncorrected two-stage benchmark, and BMLVAR under a common simulation setting.

https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html

Containers for Reproducibility

This vignette provides the containerized computing environments and step-by-step instructions needed to reproduce the analyses with the required software stack and dependencies.

https://jeksterslab.github.io/manMetaVAR/articles/containers.html

Figure 1
Overall Performance

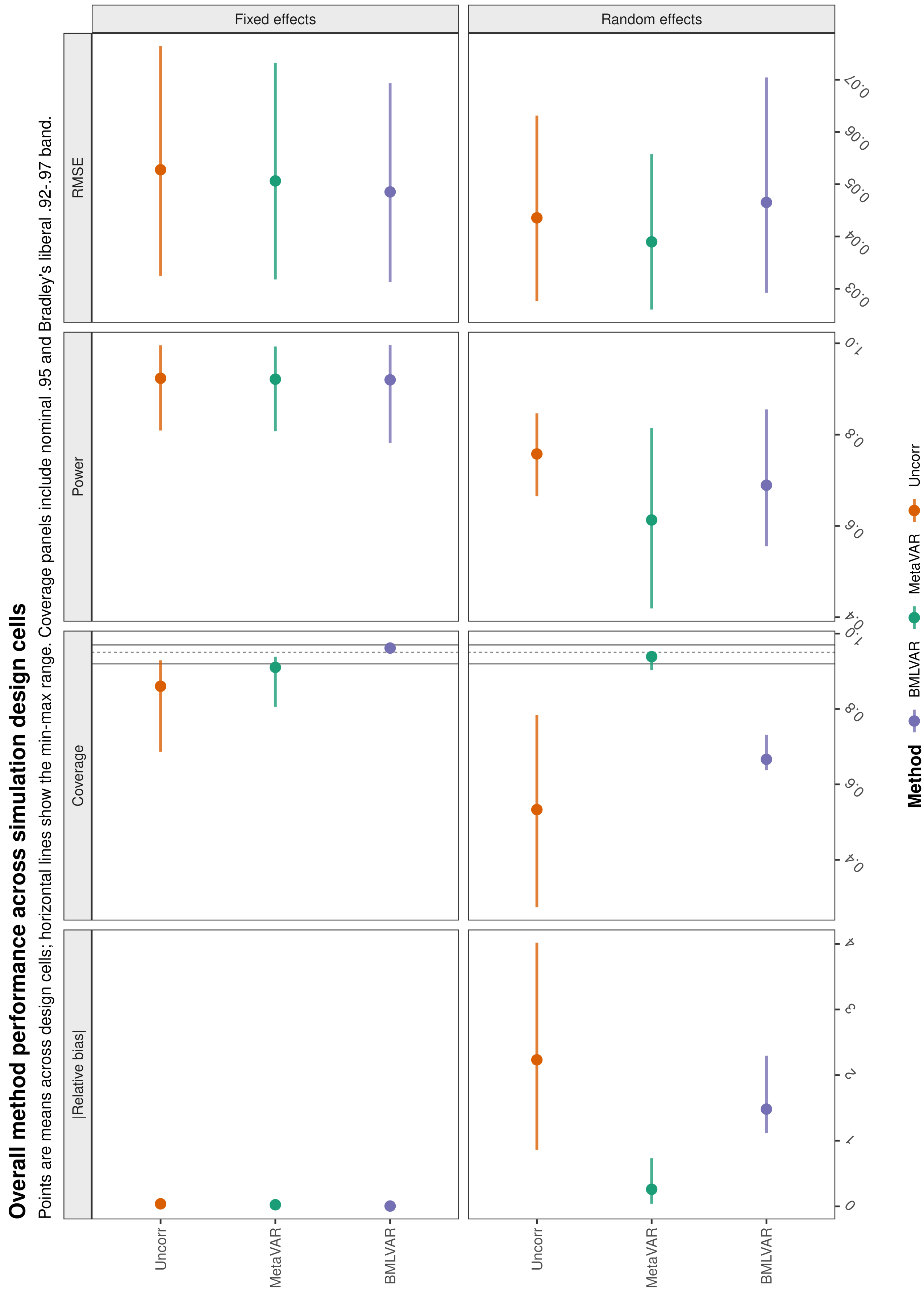

Figure 2
Coverage

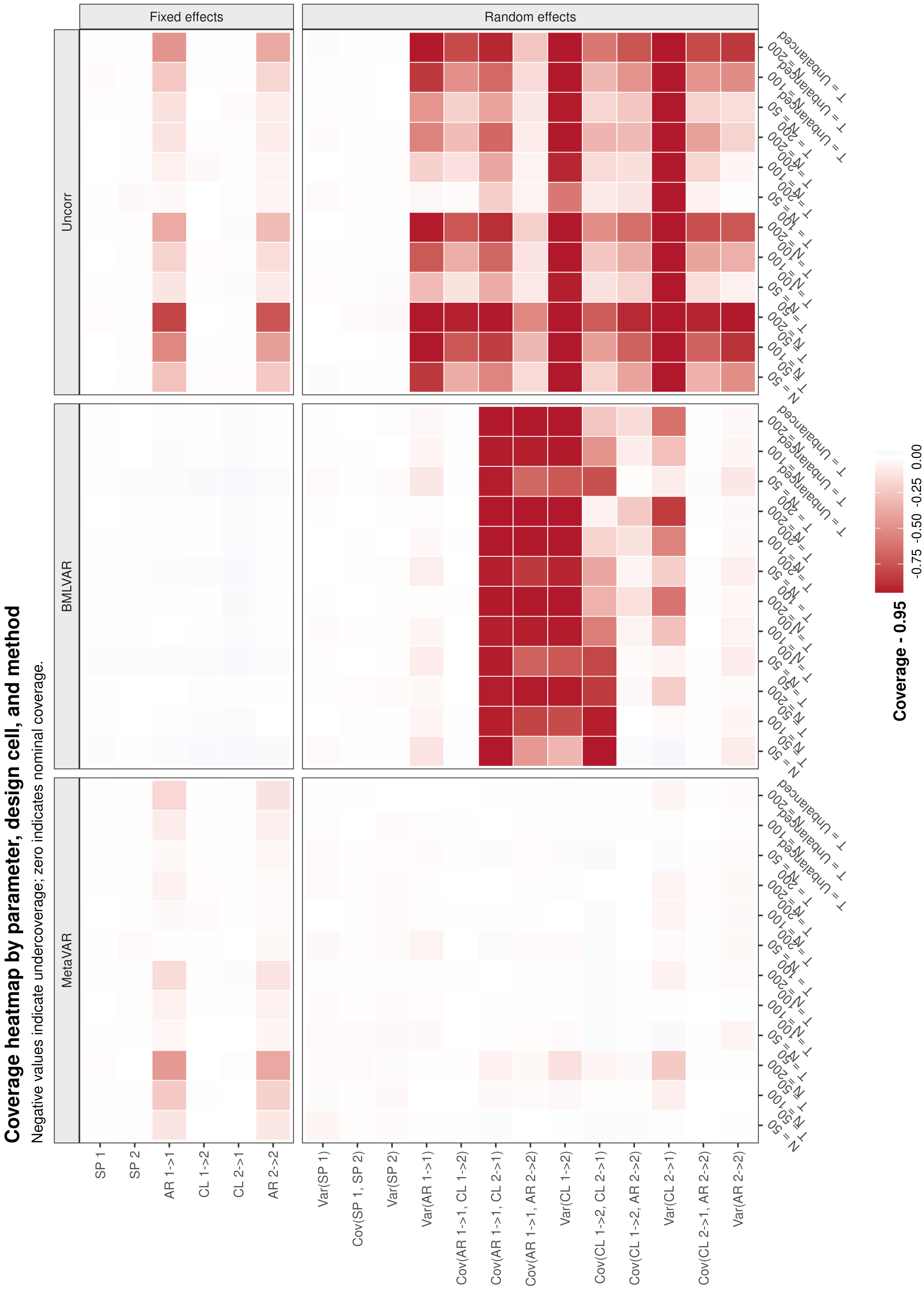

Figure 3
Relative Bias

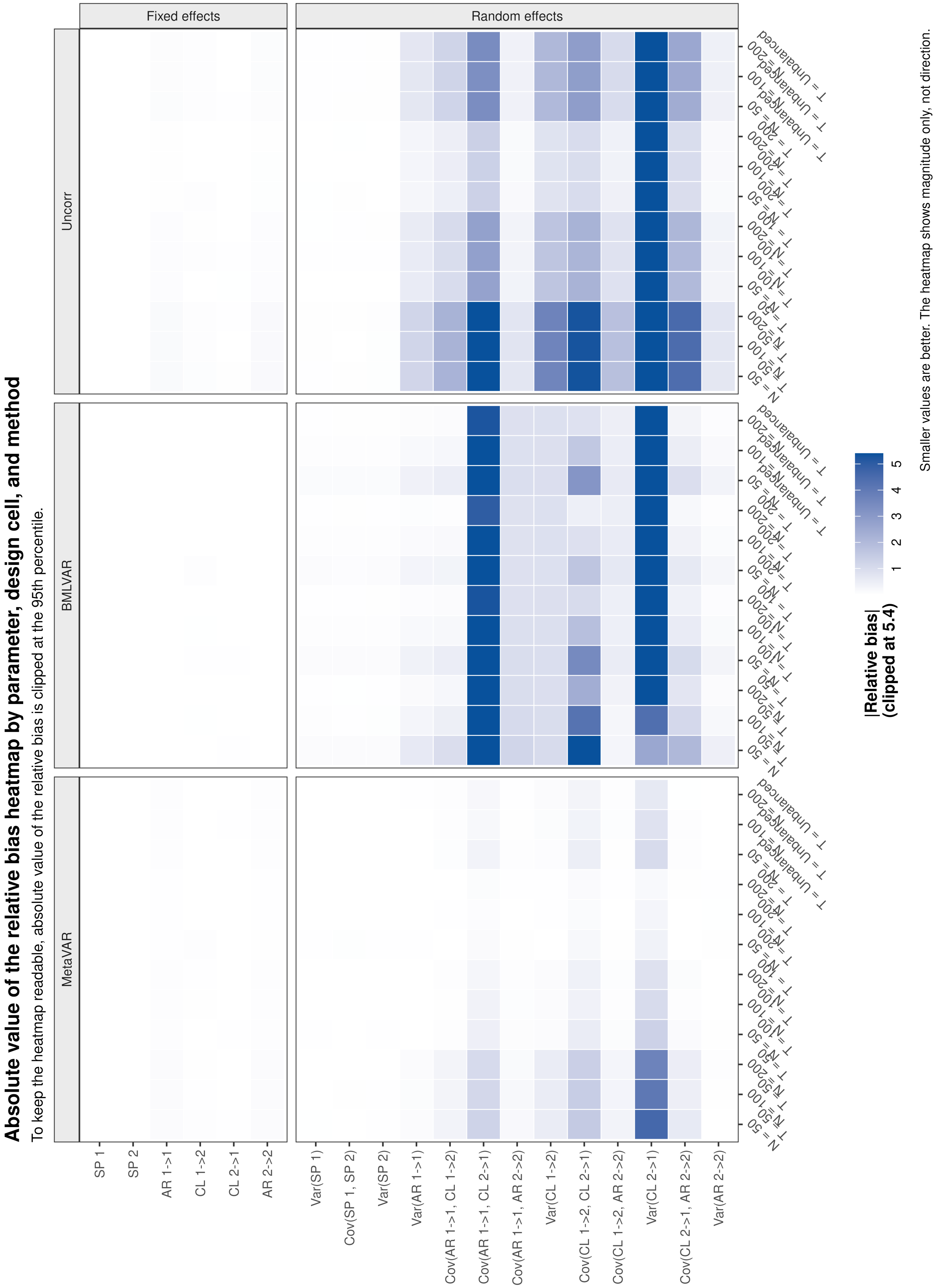

Figure 4

RMSE

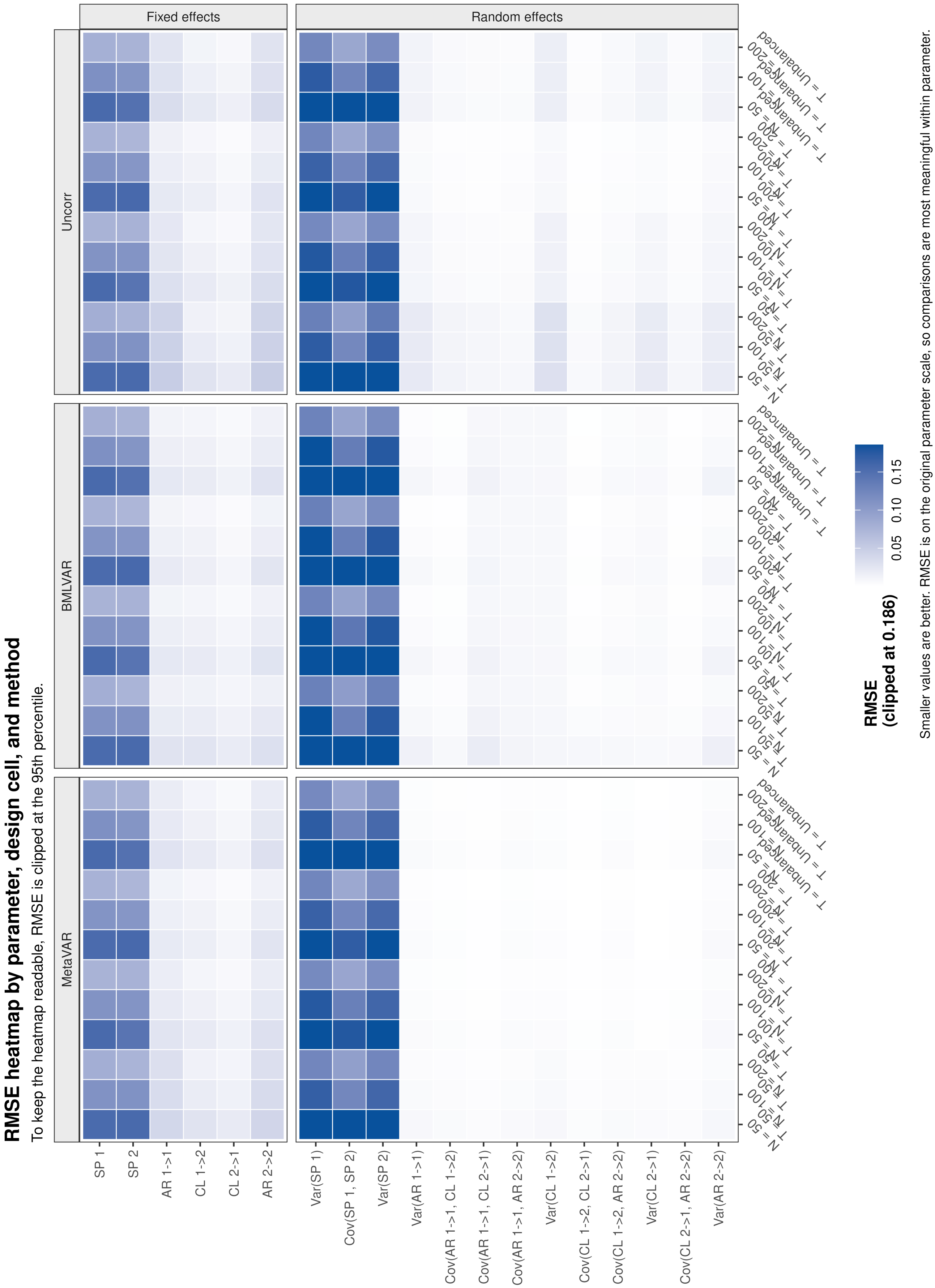

Figure 5

Power